

An oscilloscope is set with the vertical scale on 2 V/cm, and the horizontal scale on 5 ms/cm. With no input signal, the trace is in the position shown in Figure 6.

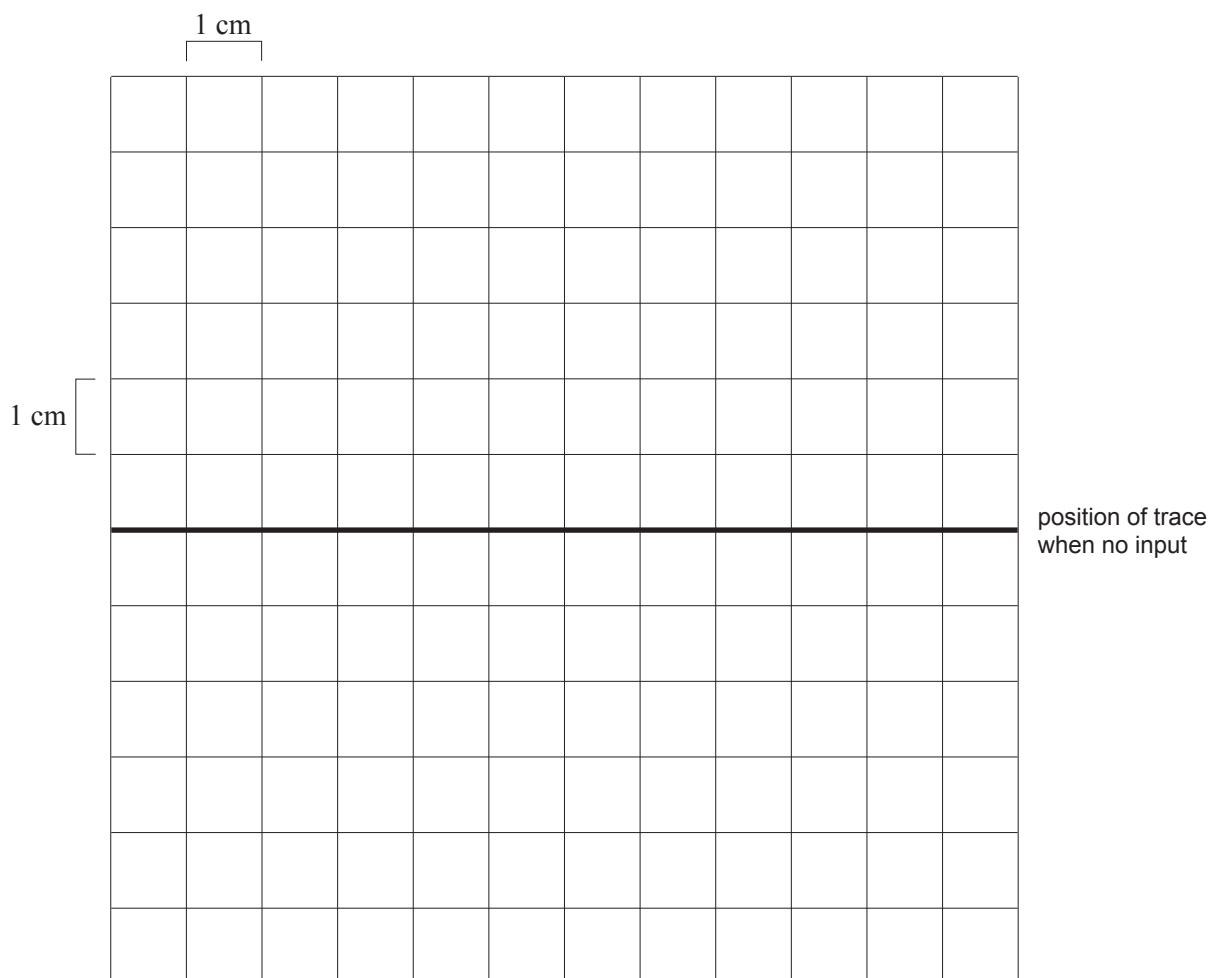


Figure 6

A $4.0 \text{ V}_{\text{RMS}}$ sine-wave signal of frequency 50 Hz is connected to the oscilloscope.

Question 12

On the diagram of the oscilloscope screen shown in Figure 6, sketch the trace that would be observed.

2 marks

PHYSICS

Written examination 1

DATA SHEET

Directions to students

Detach this data sheet before commencing the examination.

This data sheet is provided for your reference.

1	velocity; acceleration	$v = \frac{\Delta x}{\Delta t}; \quad a = \frac{\Delta v}{\Delta t}$
2	equations for constant acceleration	$v = u + at$ $x = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2ax$ $x = \frac{1}{2}(v + u)t$
3	Newton's second law	$F = ma$
4	circular motion	$a = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2}$
5	Hooke's law	$F = -kx$
6	elastic potential energy	$\frac{1}{2}kx^2$
7	gravitational potential energy near the surface of the Earth	mgh
8	kinetic energy	$\frac{1}{2}mv^2$
9	Newton's law of universal gravitation	$F = G \frac{M_1 M_2}{r^2}$
10	gravitational field	$g = G \frac{M}{r^2}$
11	stress	$\sigma = \frac{F}{A}$
12	strain	$\varepsilon = \frac{\Delta L}{L}$
13	Young's modulus	$E = \frac{\text{stress}}{\text{strain}}$
14	transformer action	$\frac{V_1}{V_2} = \frac{N_1}{N_2}$
15	AC voltage and current	$V_{\text{RMS}} = \frac{1}{\sqrt{2}} V_{\text{p-p}} \quad I_{\text{RMS}} = \frac{1}{\sqrt{2}} I_{\text{p-p}}$
16	voltage; power	$V = RI \quad P = VI$

17	resistors in series	$R_T = R_1 + R_2$
18	resistors in parallel	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$
19	capacitors	time constant : $\tau = RC$
20	Lorentz factor	$\gamma = \frac{1}{\sqrt{1 - v^2 / c^2}}$
21	time dilation	$t = t_0 \gamma$
22	length contraction	$L = L_0 / \gamma$
23	relativistic mass	$m = m_0 \gamma$
24	universal gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
25	mass of Earth	$M_E = 5.98 \times 10^{24} \text{ kg}$
26	radius of Earth	$R_E = 6.37 \times 10^6 \text{ m}$
27	mass of the electron	$m_e = 9.1 \times 10^{-31} \text{ kg}$
28	charge on the electron	$q = -1.6 \times 10^{-19} \text{ C}$
29	speed of light	$c = 3.0 \times 10^8 \text{ m s}^{-1}$

Prefixes/Units

p = pico = 10^{-12}

n = nano = 10^{-9}

μ = micro = 10^{-6}

m = milli = 10^{-3}

k = kilo = 10^3

M = mega = 10^6

G = giga = 10^9

t = tonne = 10^3 kg

END OF DATA SHEET